

DUET

A full-duplex voice agent that acts on what you mean, not on what you said first.

PLANS EARLY · COMMITS LATE · ACTS ONCE

FDB-V3 · ALL 100 REAL RECORDINGS

0.81

Pass@1, strict:
every tool, every
argument, no extras



Baselines: FDB-v3 paper. DUET: our reported run.

01 HEAR

every word advances
the epoch

02 WAIT

turn closed and the
user quiet for a hold

03 CHECK

a plan made on changed
words is dropped

04 ACT ONCE

the ledger never runs
an action twice

05 ANSWER

spoken from the tool
results, never guessed

Team SE7EN · SRM Institute of Science and Technology · SRM_SE7EN

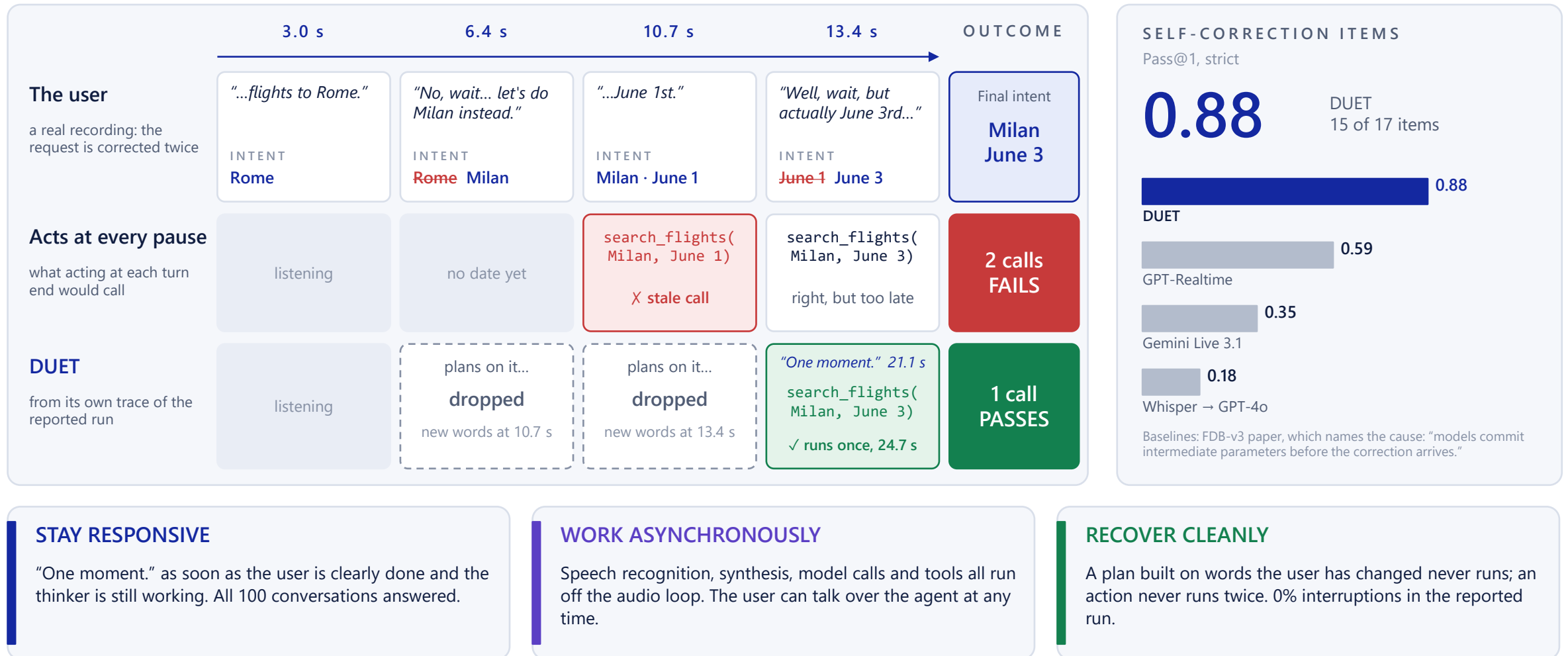
Tanmay Singh · Pranjal · Naman

Gemma 4 26B-A4B · LiveKit Agents · faster-whisper · Kokoro-82M · one command to reproduce

github.com/tannmayy/DUET-SAMSUNGPRISM-HACKATHON-TEAM-SE7EN

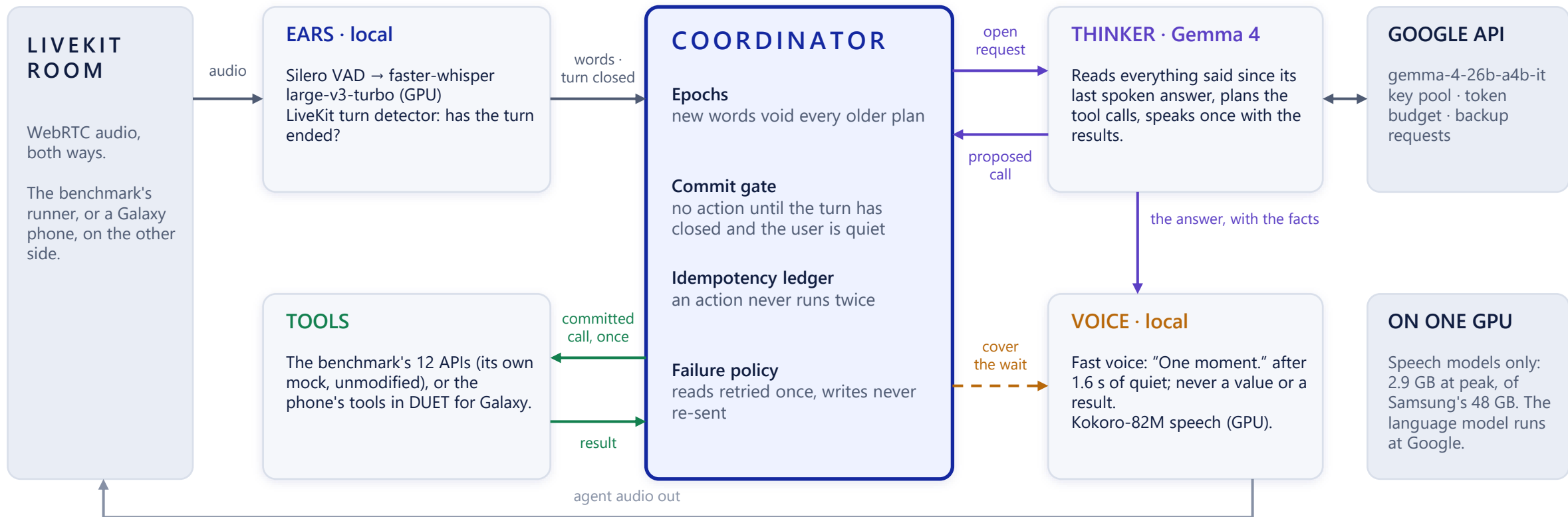
The failure isn't hearing the correction. It's acting before it arrives.

One real Full-Duplex-Bench recording (travel_19), moment by moment. Times are from DUET's own trace of the reported run.



Two minds and a referee: every action leaves through one guarded exit

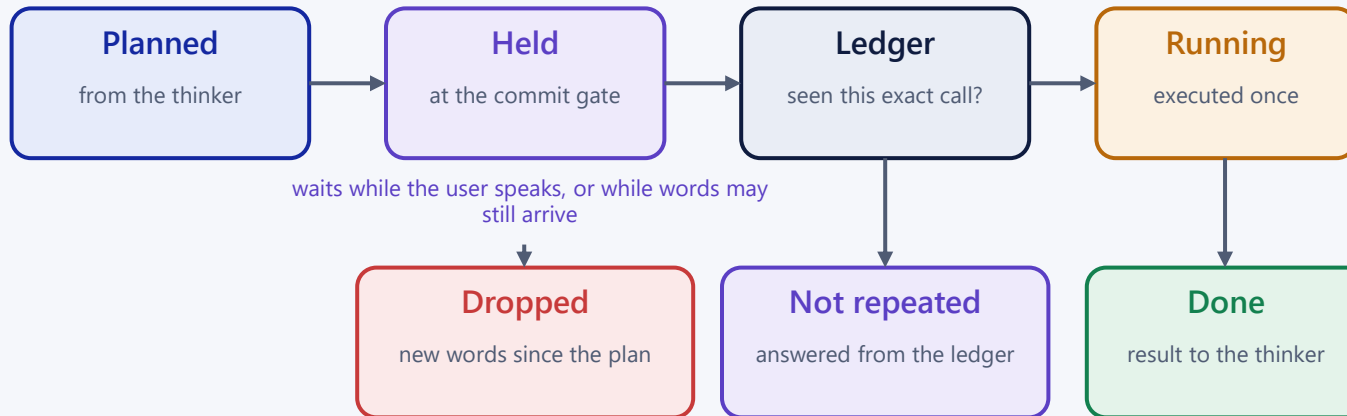
The model decides what should happen. Deterministic code decides whether, and when, it may happen.



Every tool call in the benchmark and on the phone takes this one path, so the same guarantees hold everywhere: a correction always wins, and nothing is done twice.

Plans early, commits late, acts once

Each proposed action travels a small state machine. Only the path to Done ever touches the outside world.



Running can also end in:

A read failed
retried once, then the thinker is told

A write timed out
outcome unknown: never re-sent; the user is told and offered a human

Epochs

Every plan carries the epoch of the words it was made on. New words void it, even after the newer turn has closed. A cough brings no words: it delays an action, never cancels it.

Commit gate

A tool runs only after the turn closes and the user is quiet for 1.1 s; 1.8 s after a revision ("no wait"); 2.2 s when the sentence is left open ("and...").

Idempotency ledger

An identical call already made in the conversation is answered from the ledger, so a booking survives a barge-in and never happens twice.

Keep listening

"The order number is...": the thinker says nothing and does nothing until the user has been quiet for 2.5 s.

Why it matters for Pass@1: the score is strict, so one stale or duplicated call fails the item. In the reported run, 33 model calls were abandoned mid-flight because the user's new words had already overtaken their plan.

Built to survive someone else's machine and someone else's API key

Samsung re-runs our script; only that run counts. So every model call is defended, and every version is pinned.

EVERY MODEL CALL · `duet_voice/gemma_api.py`

- 1 Book the call on the key with the most room: 15,000 of Google's 16,000 input tokens a minute per key
- 2 Rate-limited (429): rest that key for the delay Google asks, try the next
- 3 Server error (500, 503, 504): retry in 0.3 to 0.7 s
- 4 No answer after 6 s: send a backup to another key (at most 4 in flight); the first answer wins
- 5 28 s deadline per call, inside the recording's 30 s window

IN THE REPORTED RUN

220

calls answered

3.3 s

median answer

221transient errors
retried**109**

backups sent

TECH STACK

Voice framework

LiveKit Agents 1.8.3 (custom agent)

Language model

Gemma 4 26B-A4B-it · Google API · temperature 1.0,
top-p 0.95, top-k 64, seed 7

Speech in

faster-whisper large-v3-turbo @ 0a363e9 · Silero VAD ·
LiveKit turn detector

Speech out

Kokoro-82M @ f3ff357, voice af_heart

Benchmark

Full-Duplex-Bench v3 @ 3e799c4: runner, mock tools
and evaluations unmodified

Extension

Capacitor Android app · LiveKit Cloud · RPC duet.tool

ONE COMMAND

```
bash reproduce.sh
```

- Installs Python 3.11 and every package from lock files (uv)
- Benchmark commit, data SHA-256, LiveKit checksum pinned
- Preflight tests every key before anything runs
- Official runner, then the three official evaluations
- Dockerfile runs the same script

TRUST

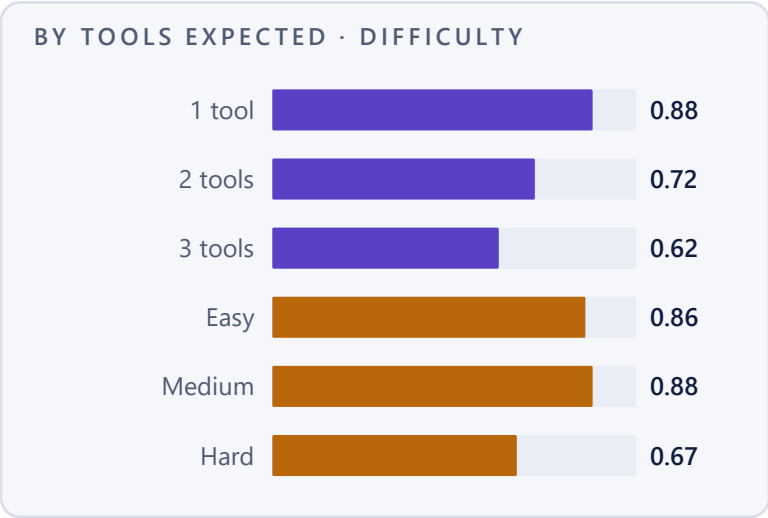
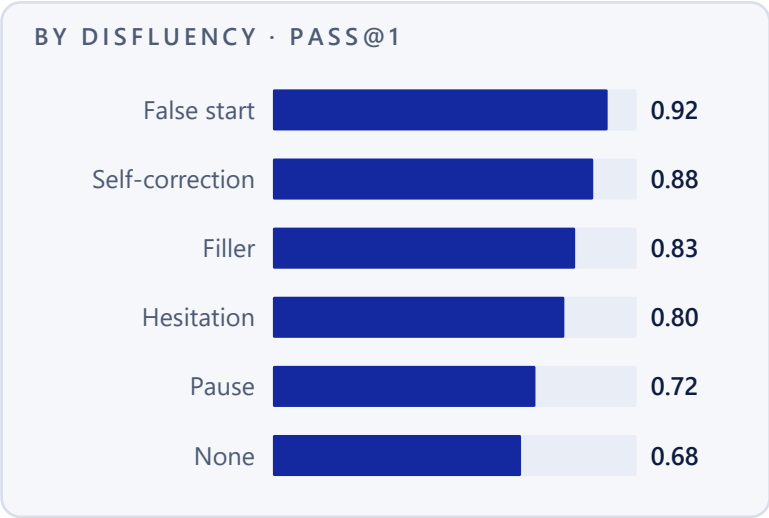
- 75 unit tests
- A test fails if any benchmark answer appears in the agent
- A trace of every conversation; one log line per model call

Full-Duplex-Bench v3: 0.81 Pass@1 on all 100 real recordings

Streamed in real time through the benchmark's own runner; scored by its own evaluation scripts.

System	Pass@1	Tool F1	Arg. acc.	Response	Turn-taking	Key info	Interrupts
DUET · Gemma 4 26B-A4B	0.81	0.915	0.850	0.730	100%	14.4 s	0%
GPT-Realtime	0.600	0.876	0.680	0.792	96.0%	6.9 s	13.5%
Gemini Live 3.1	0.540	0.817	0.588	0.718	78.0%	4.3 s	19.2%
Whisper → GPT-4o → TTS	0.450	0.803	0.562	0.600	100%	10.1 s	33.0%

Green: best in its column. Baselines: FDB-v3 paper. Response: response quality. Key info: time to the answer's key facts.



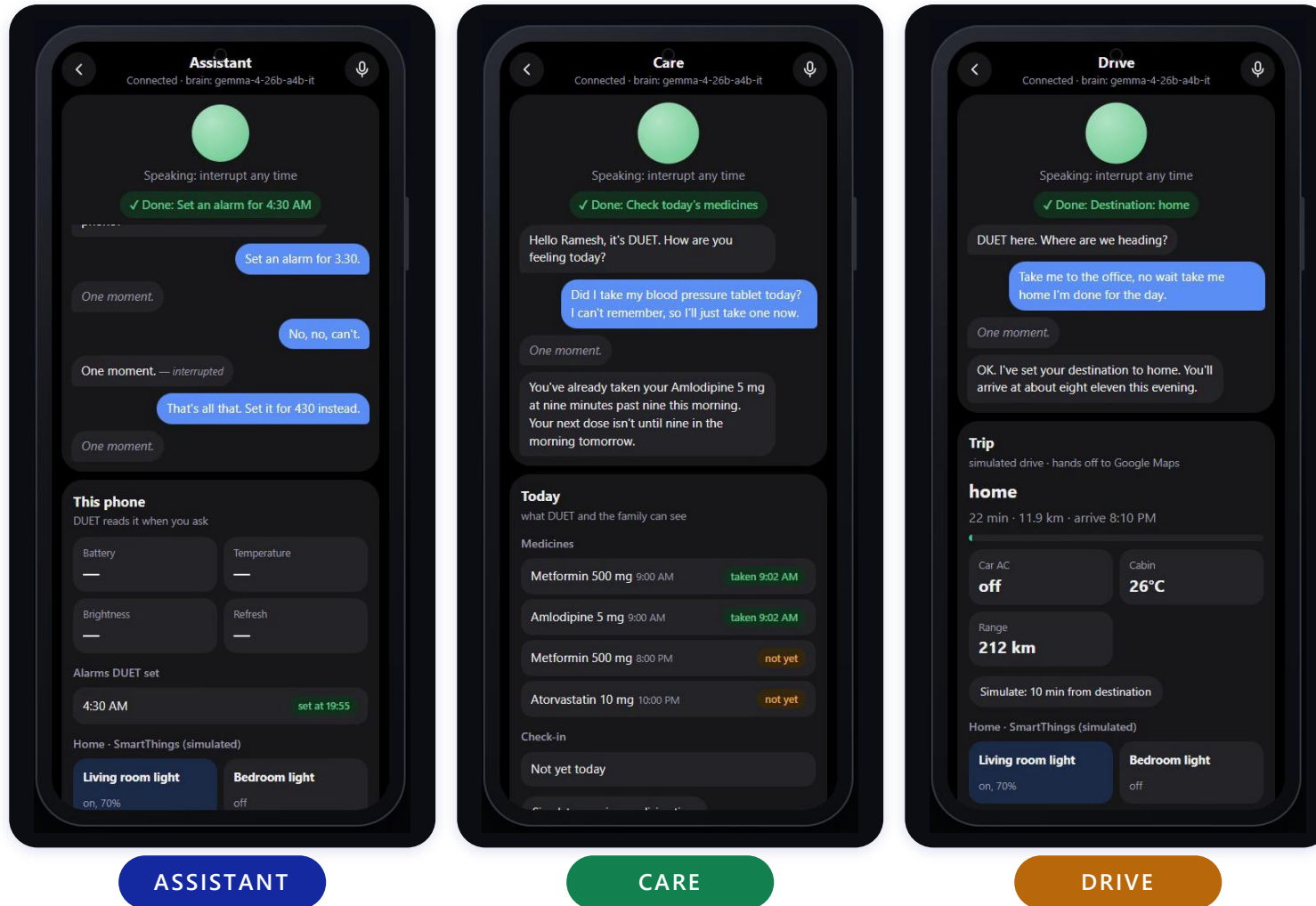
THE 25 EXACT-MATCH MISSES

- 8 **Format only**
right tool and value, other wording
- 7 **Heard wrong**
spelled codes the recognizer misheard
- 3 **Provider stalls**
Google answered nothing for 28 s
- 3 **Conditional**
expected answer contradicts the mock data
- 2 **Not passable**
expected value never spoken, or not the one said
- 2 **Judgement**
DUET asked instead of acting

How it was scored. The judge was Gemma 4 standing in for gpt-4o, with the official prompts unchanged. With no judge at all, by exact match, DUET passes 75 of 100. Samsung's re-run uses gpt-4o.

The same agent, coordinator and model on a Samsung phone

An Android app in three modes. Real runs on Gemma 4, recorded; every phone action passes the same coordinator.



ASSISTANT "Set an alarm for 3:30... no, set it for 4:30."

The 3:30 plan never runs. One alarm, 4:30.

CARE "Did I take my tablet? I'll just take one now."

DUET checks the record and gently stops a second dose.

DRIVE "Take me to the office, no wait, home."

Only home is set; the ETA text to Priya goes once.

HOW IT WORKS

Phone $\rightleftharpoons$ LiveKit Cloud $\rightleftharpoons$ DUET agent $\rightleftharpoons$ Gemma 4. The app shows each step live: Planned, Never ran, Done, Not repeated.

Real Android interfaces: alarms, battery and screen readings, apps, brightness, settings, Maps, calls, messages. Simulated in the app: the SmartThings home, the medicine schedule, the car.

8 of 8 phone scenarios pass end to end
(scripts/galaxy_check_scenarios.py).

Recorded on the app's web build with a scripted user voice. The agent, the model and every tool call are live; the web build simulates the phone's side, which Android carries out on a Galaxy.

What we still lose, and where DUET goes next

Every limitation below is measured in the reported run; every next step is a PRISM worklet.

LIMITATIONS WE MEASURED

Latency. First response 4.8 s; key information 14.4 s (GPT-Realtime: 6.9 s). A hosted model call takes 3.3 s at the median, 7.2 s at p90.

Provider stalls. 3 of 100 recordings were lost to model calls Google never answered (504), despite backups on every key.

Hearing. Spelled-out codes are misheard ("D, E, L, bye, V" for DELIV): 7 of 100 items.

Response quality 0.73 (GPT-Realtime 0.79): the answers are correct but plain.

Our scoring. A Gemma stand-in for the gpt-4o judge; 75 of 100 by exact match with no judge at all.

The extension's demo runs the app's web build; the home, medicines and car are simulated.

NEXT, AS A PRISM WORKLET

- 01 On-device Gemma on Galaxy**
No round trip and no key limits: the answer's latency drops, and the user's words never leave the phone.
- 02 Confirm before acting on codes**
Recognizer confidence per word; a low-confidence order id is read back before the tool runs.
- 03 Speculative reads, gated writes**
Start safe lookups while the user is still talking; keep every state change behind the gate.
- 04 SmartThings and Android Auto**
The Care and Drive modes on real devices, with Knox-consented health reads.
- 05 Field metrics**
Stale-action rate, duplicate-action rate and time to first feedback, measured on real users.

ROUND 2: INTERRUPT IT LIVE

Reproduce: `bash reproduce.sh` The app: `app/README.md` Every trace: `results/reported/`